

व्यावसायिक परीक्षण रिपोर्ट (प्रारंभिक)

**COMMERCIAL TEST REPORT
(INITIAL)**

संख्या /No: Machine-422/1256

माह:/Month: March, 2021

Validity: 28.02.2026



FARMWORLD FWMT420DE POWER WEEDER



भारत सरकार

Government of India

कृषि एवं किसान कल्याण मंत्रालय

Ministry of Agriculture and Farmers Welfare

कृषि, सहकारिता एवं किसान कल्याण विभाग

Department of Agriculture, Cooperat on and Farmers Welfare

दक्षिणी क्षेत्र कृषि मशीनरी प्रशिक्षण एवं परीक्षण संस्थान

Southern Region Farm Machinery Training and Testing Institute

ट्रैक्टर नगर, गार्लदिन्ने-515 731, जिला: अनंतपुर (आं. प्र.)

Tractor Nagar, Garladinne-515 731, District: Anantapur (A.P.)

[An ISO 9001:2015 CERTIFIED INSTITUTE]

Website: <http://srfmtti.dacnet.nic.in/>

E-mail: fmti-sr@nic.in

Tele.: 08551-286441

Machine- 422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
------------------------------	---	-----------------------------

Applicant : M/s. Farmworld Agrotech Pvt. Ltd.,
SF 08, Shreemad Bhavan, Opp. Kanta
Stree Vikas Gruh, Dhebar Road,
Rajkot – 360 002, Gujarat, **India.**

FARMWORLD FWMT420DE POWER WEEDER

Report No. : Machine-422/1256 Month: March Year: 2021



सत्यमेव जयते

Government of India
Ministry of Agriculture and Farmers Welfare
Department of Agriculture, Cooperation and Farmers Welfare
Southern Region Farm Machinery Training and Testing Institute
Tractor Nagar, Garladinne-515 731, District: Anantapur (A.P.)
[An ISO 9001:2015 CERTIFIED INSTITUTE]

Website: <http://srfmtti.dacnet.nic.in/>

E-mail: fmti-sr@nic.in

Telephone: 08551-286441

Machine-422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
-------------------------	---	-----------------------------

Type of test : INITIAL COMMERCIAL TEST

Period of test : 08.02.2021 to 03.03.2021

Test Report No. : Machine-422/1256

Month / Year of release : March, 2021

- i) The results reported in this report are observed values and no corrections have been applied for atmospheric and site conditions.
- ii) The data given in this report pertain to the particular machine submitted by the applicant.
- iii) The results presented in this report do not in any way attribute to the durability of the machine.
- iv) This Test Report should not be reproduced in part or full without prior permission of the Director, Southern Region Farm Machinery Training & Testing Institute, Garladinne, Anantapur (A.P.).
- v) This is a report on Commercial Test of Power Weeder named "Farmworld FWMT420DE Power Weeder " **This report is valid up to 28.02.2026, Vide Ministry's O.M. No. 13-24/2018-M&T (I&P) dated 19.09.2018.**

SELECTED CONVERSIONS

S. No.	Units	Conversion Factor
1	Force	
	1 kgf	9.80665 N
		2.20462 lbf
2	Power	
	1 hp	1.01387 metric hp (Ps)
		745.7 W
	1 Ps	735.5 W
	1 kW	1.35962 Ps
3	Pressure	
	1 psi	6.895 kPa
	1 kgf/cm ²	98.067 kPa = 735.56 mm of Hg
	1 bar	100 kPa = 10 N/cm ²
	1 mm of Hg	1.3332 m-bar

Machine- 422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
------------------------------	---	-----------------------------

Type of test : INITIAL COMMERCIAL TEST

Name of machine : Power weeder

Make (apa) : Farmworld

Model : FWMT420DE

Type : Self-propelled, Walk behind

Applicant : M/s. Farmworld Agrotech Pvt. Ltd.,
SF 08, Shreemad Bhavan, Opp. Kanta Stree
Vikas Gruh, Dhebar Road,
Rajkot – 360 002, Gujarat, **India.**

TESTED BY

Government of India
Ministry of Agriculture and Farmers Welfare
Department of Agriculture, Cooperation and Farmers Welfare
Southern Region Farm Machinery Training and Testing Institute
Tractor Nagar, Garladinne-515 731, District: Anantapur (A.P.)
[An ISO 9001:2015 CERTIFIED INSTITUTE]

Machine- 422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
------------------------------	---	-----------------------------

C O N T E N T S

Sl. No.	Chapters	Page No.
1	Scope of Test	6
2	Method of Selection	6
3	Test Code/Test Procedure	6
4	Specifications	7
5	Fuel & Lubricants	16
6	Running-in	16
7	Engine Performance Test	17
8	Amplitude of Mechanical Vibration	22
9	Noise Measurement	22
10	Air cleaner oil pull over test	23
11	Turning Ability	23
12	Hardness and Chemical Composition	24
13	Field Performance Test	25
14	Ease of Operation and Adjustments	26
15	Defects, Breakdowns and Repairs	26
16	Components / Assembly Inspection	26
17.	Comments and Recommendations	29
18.	Applicant's comments	30
	Annexure	31

Machine- 422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
------------------------------	---	-----------------------------

1. SCOPE OF TEST

The scope of test was limited to check and assess the following:

- 1.1 Specification and other data furnished by the applicant.
- 1.2 Engine performance test
- 1.3 Amplitude of mechanical vibration
- 1.4 Noise measurement
- 1.5 Air cleaner oil pull over test
- 1.6 Turning ability
- 1.7 Hardness & Chemical composition
- 1.8 Field performance
- 1.9 Wear analysis of rotor blades
- 1.10 Ease of operation and adjustments
- 1.11 Defects, breakdowns and repairs

2. METHOD OF SELECTION

The test sample was directly submitted by the applicant for test at this Institute. Hence the method of selection is not known.

3. TEST CODE/TEST PROCEDURE

There is no Indian standard/test code available for testing of self propelled power weeder as such. The guidelines, however, have been taken from the following:

- | | |
|-------------------------------------|---|
| IS 9935 : 2002
(Reaffirmed 2012) | : Power Tiller - Test code |
| IS 9980 : 1999
(Reaffirmed 2004) | : Guidelines for field performance and haulage tests of power tillers |
| IS 1976 : 1976
(Reaffirmed 2009) | : Specification for Rotary paddy weeder, manually operated |
| IS 6690 : 1981
(Reaffirmed 2012) | : Specification for Blades for Rotavator for Power Tillers |

Machine- 422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
------------------------------	---	-----------------------------

4. SPECIFICATION

4.1 General:

Make (apa)	: Farmworld
Model	: FWMT420DE
Name of applicant	: M/s. Farmworld Agrotech Pvt. Ltd., SF 08, Shreemad Bhavan, Opp. Kanta Stree Vikas Gruh, Dhebar Road, Rajkot – 360 002, Gujarat, India.
Name of machine	: Power weeder
Type of machine	: Self propelled, walk behind
Indian trade name	: Mini tiller
Working width of machine, (mm)	: 1450
Year of manufacture	: 2020
Serial no. of machine	: 2020211308

4.2 Details of prime mover:

Make	: Farmworld
Model	: LA186FAE
Type	: Vertical, 4 stroke, Air cooled, Compression Ignition Engine Direct ignition
Year of manufacture	: 2020
Engine Serial No.	: F13BJJH0014
Recommended high idle speed, rpm	: 3880 ± 100
Recommended low idle speed, rpm	: 1300
Recommended rated speed, rpm	: 3600
Speed at maximum torque, rpm	: 2880
Recommended rated engine speed for field, rpm	: 3600
Max. power observed, kW (metric hp)	: 6.4 (8.7) @ 3600 rpm

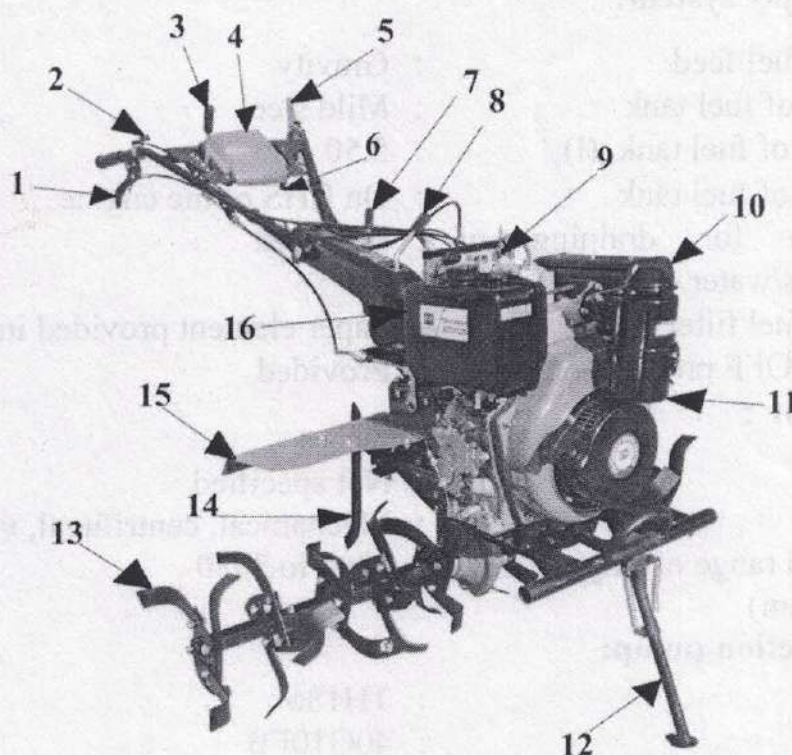


Fig.1 Farmworld FWMT420DE Power Weeder

KEYS:

- | | |
|--|----------------------------|
| 1. Reverse gear engaging/disengaging lever | 9. De-compression lever |
| 2. Accelerator lever | 10. Silencer |
| 3. Main gear shifting lever | 11. Air cleaner |
| 4. Toolbox | 12. Parking stand |
| 5. Emergency stop lever | 13. Rotary blade |
| 6. Main clutch lever | 14. Depth adjustment lever |
| 7. Handle height adjustment lever | 15. Rotary cover |
| 8. Handle rotation adjustment lever | 16. Fuel tank |

4.2.1 Cylinder and cylinder head:

Number	: One
Disposition	: Vertical
Bore/Stroke, mm (apa)	: 86/72
Capacity, cc	: 418
Compression ratio (apa)	: 19±1:1
Valve clearance, (mm) (cold)	

Inlet	: 0.15
Exhaust	: 0.15

Cylinder liner	: Dry, Inbuilt
Valve arrangement	: Over head valve

Machine- 422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
------------------------------	---	-----------------------------

4.2.2 Fuel supply system:

Type of fuel feed	: Gravity
Material of fuel tank	: Mild steel
Capacity of fuel tank, (l)	: 5.50
Location of fuel tank	: On RHS of the engine
Provision for draining of sediments/water	: Provided
Type of fuel filter	: Paper element provided in fuel tank
Fuel ON/OFF provision	: Provided

4.2.2.1 Governor :

Make	: Not specified
Type	: Mechanical, centrifugal, variable speed.
Governed range of Engine speed, (rpm)	: 1300 to 3980

4.2.2.2 Fuel injection pump:

Make	: TH186
Model	: 40G10FB
Type	: Plunger & barrel
Method of drive	: Through camshaft

4.2.3 Fuel injector :

Make	: TH186FA
Model	: Not specified
Type	: Nozzle element, Multi hole
Manufacturer's recommended pressure setting, (kgf/cm ²) (apa)	: 204±5
Injection timing (apa)	: 19 to 21° before TDC

4.2.4 Air-intake system:

4.2.4.1 Pre-cleaner : Not provided

4.2.4.2 Air-cleaner:

Make	: Not specified
Type	: Wet type
Type of element	: Wire mesh
Location of air-cleaner	: In front of the engine
Capacity of air cleaner bowl (apa), (l)	: 0.040
Recommended service schedule	: After every 100 hours of operation
Recommended grade of oil	: SAE 10W30

Machine- 422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
------------------------------	---	-----------------------------

4.2.5 Exhaust:

Type	: Side draft, Rectangular
Location	: LHS of the engine
Provision of cover on exhaust	: Provided
Direction of exhaust emission away from operator	: Provided

4.2.6 Lubrication system:

Type	: Force feed cum splash
Type of lubricant recommended	: SAE 10W30
Oil capacity, (l)	: 1.65
Type of pump	: Gear type
No. of oil filters	: One
Permissible oil pressure (kgf/cm ²) (apa)	: ≥ 0.08
Relief valve pressure setting (Kgf/cm ²) (apa)	: Not specified
Oil change period	: First after 50 hours then subsequently after every 100 hours or 3 months of operation

4.2.7 Cooling system:

Type	: Air cooled
------	--------------

4.2.7.1 Fly wheel and fins details:

Details:

Material of fins of flywheel	: Cast iron in built of fly wheel
No. of fins (Ribs)	: 27
Height x Width x Thickness of fins, (mm)	: 29.31x40.76x5.19
Dia. of fly wheel, (mm)	: 270.30
Width of fly wheel with fins, (mm)	: 74.30

4.2.8 Starting system :

Type	: Manual, recoil and self start
Ignition system	: Compression ignition
Aid for cold starting	: None
Any other provision for easy starting	: De-compression lever provided

4.3 Electrical system:

4.3.1 Battery:

Make	: Amaron lead acid
Model	: Quanta
Serial No.	: BAQ01851U100395

Machine-422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
-------------------------	---	-----------------------------

	Type	: Lead acid (wet type)
	Capacity and rating	: 12V 18 A.H.
4.3.2	Starter:	
	Make	: Not specified
	Model	: 186FAE general purpose
	Serial No.	: 20200718LA188
	Type	: Solenoid type
	Capacity & rating	: Output 100W
4.4	Power transmission system:	
4.4.1	Clutch	
	Make	: Not specified
	Type of clutch	: Wet type
	Dia. of clutch plate, (mm)	: Ø 110.38
	No. of friction plates	: 05
	No. of pressure plates	: 06
	Material of lining	: Not specified
	No. of coil springs	: 04
	Location of clutch	: Inside the gear box
4.4.2	Primary (Main) gear box:	
	Make (apa)	: Farmworld
	Type	: Sliding and constant mesh
	No. of speeds	: 03, (Forward - 02 ; Reverse - 01)
	Grade of oil	: SAE 90
	Oil capacity, (l)	: 2.0
	Oil changing period	: After every 50 hours of operation
	Reduction ratio :	
	Forward	
	1 st gear	: 2.786:1
	2 nd gear	: 1.41:1
	Reverse	: 3.718:1
	No. and size of bearing	: 05 Nos. (03 Nos. of ball bearings, 1 thrust bearing, 1 needle bearing)
4.4.3	Secondary reduction gear box for rotary shaft:	
	Make (apa)	: Farmworld
	Type	: Constant mesh (Bevel gears)
		1 st Pair 2 nd Pair
	No. of teeth on Pinion	: 10 10
	No. of teeth on bevel gear	: 43 36
	Reduction ratio	: 4.3:1 3.6:1

Machine- 422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
------------------------------	---	-----------------------------

Overall reduction ratio from engine crankshaft to rotary shaft

Forward

1st gear : 43.13:1

2nd gear : 21.83:1

Reverse : 57.55:1

No. of bearing

: 05 Nos. (02 Nos. taper roller bearings 32009 & 30204), (03 Nos. ball bearing 6009, 6202 & 6204)

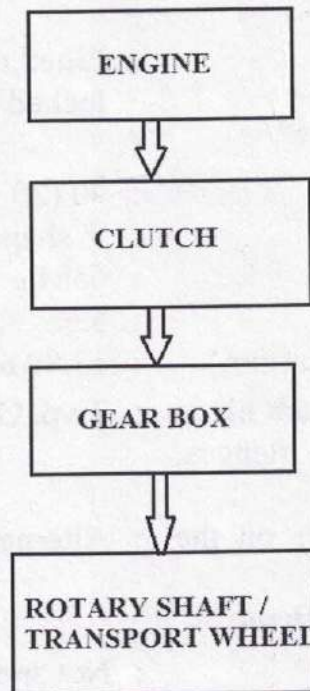


Fig. 2 Schematic power flow diagram

4.5 Rotary unit:

Make : Farmworld

4.5.1 Rotary cover :

Material : Mild steel

Size, (mm) : 370x245

Sheet thickness, (mm) : 1.40

Method of fixing : A frame is made of MS angle iron of size 28.95x3.12 (t) mm used to bear the rotary cover of 245x3.7x1.40 (t) mm

4.5.2 Rotary shaft:

Material : Not specified

Type of rotary axle : Hexagonal

Size of shaft, mm : 395 x Ø36.10

Machine-422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
-------------------------	---	-----------------------------

No. of flanges	: 10(5+5)
Type of flanges	: Square
Size of flanges, (mm)	: 97.27/97.27
Thickness of flanges, (mm)	: 4.87
No. of blades on each flange	: 04
Method of mounting blades on flanges	: Bolted
Distance between two consecutive flanges, (mm)	: 130
Dia. of rotary with blades, (mm)	: 345
Method of fixing	: Fitted rotary with the help of cotter pin & locked with the help of split pin

4.5.3 Rotary blades:

Numbers	: 40 (20 + 20)
Shape of blade	: 'J' shape
Material (apa)	: 65Mn
Thickness, (mm)	: 5.56
Width of beveled edge, (mm)	: 13.89 to 17.92
No. & size holes on each blade for fixing it to the flanges, (mm)	: Two, Ø 10.92
Arrangement of blades on the axle	: Alternately on LHS and RHS

4.6 Details of transport wheel:

Make of tyre	: Not specified
Number & Type	: Two, Pneumatic, Traction
Size	: 5.00-12; 6PR
Max. loading capacity, (kg)	: 255 kg at 200 kPa
Method of fixing	: Fitted with help of cotter and shaft pin on rotor axle.

Machine- 422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
----------------------	-------------------------------------	---------------------

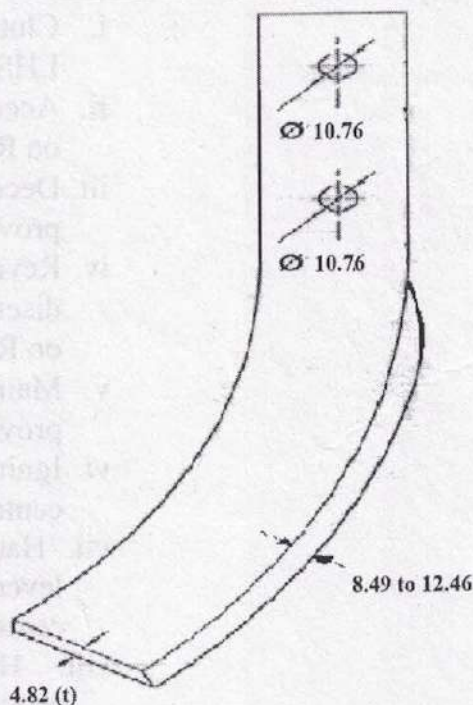


Fig. 3 Rotary blade
(All dimensions in mm)

4.7 Depth control mechanism:

Type & material	:	MS flat
Size of flat, (mm)	:	
- Length (curved)	:	470
- Width	:	38.55
- Thickness	:	11.73
No. of flats	:	One
Provision for depth adjustment	:	08 nos. of holes are provided

4.8 Steering handle bar:

No. of steering handle	:	Two
Material	:	Mild steel
Dia. of pipe, (mm)	:	25.30
No. of hand grips	:	Two
Material of grip	:	Rubber
Length of grip, (mm)	:	122.09
Dia. of grip, (mm)	:	39.26
Height of handle bar from ground level, (mm)	:	850- 1610
Provision for handle height adjustment	:	Provided
Provision for handle rotation	:	Provided
Provision for rear PTO	:	Provided

Machine- 422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
------------------------------	---	-----------------------------

- 4.9 Controls :**
- Clutch lever is provided on LHS of handle bar.
 - Accelerator lever is provided on RHS of handle bar.
 - Decompression lever is provided on engine
 - Reverse gear engaging and disengaging lever is provided on RHS of handle bar.
 - Main gear shifting lever is provided on RHS of handlebar
 - Ignition switch is provided at centre of handle bar
 - Handle height adjustment lever is provided at LHS of the centre handle bar
 - Handle rotation lever is provided on center of handle bar.
 - Provision for PTO is provided at rear of gearbox housing.
 - Emergency stop lever is provided at LHS of handlebar

4.10 Marking/labeling:

A metallic labeling plate is riveted on handlebar base with the following details:

FW Honour Your Trust Mini Tiller		
Model	:	FWMT420DE
Engine	:	4 stroke, Diesel
Rated power	:	5.9kW/3600 rpm
Displacement	:	418
Tilling depth	:	100 - 250 mm
Tilling width	:	750 - 1350 mm
Serial no.	:	2020211308
Year	:	2020
Farwmworld Agrotech Private Limited, Rajkot, Gujarat, India www.farmworld.in		

Machine-422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
-------------------------	---	-----------------------------

4.11 Overall dimensions, (mm):

Length	: 1600
Width	:
-With transport wheel	: 985
-With rotary unit	: 1460
Height	: 1330

4.12 Mass, (kg) :

With rotary unit	: 138.0
With transport wheel	: 141.5

4.13 Colour of machine:

Engine	: Silver
Chassis, silencer, Air cleaner, handlebar & fuel tank	: Black
Main drive cover, Rotary cover, fly wheel cover & wheel disc	: Green

5. FUEL & LUBRICANTS

5.1 Fuel : The high-speed diesel oil supplied by M/s Hindustan Petroleum Corporation Limited having density of 0.825g/c.c. at 15⁰ C was used.

5.2 Lubricants

Sl. No.	Particulars	As recommended by the manufacturer	As used during the test
1	Engine	SAE 10W30	As per recommended
2	Air cleaner	SAE 10W30	
3	Gear box	SAE 90	

6. RUNNING-IN

The power weeder was Run-in for 1.0 hours before the actual test. All the fasteners were checked and tightened thereafter.

Machine- 422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
----------------------	-------------------------------------	---------------------

7. ENGINE PERFORMANCE TEST

Date(s) of test : 24.02.2021, 25.02.2021 and
26.02.2021

Power Weeder run at the Institute : 34.50
before engine performance test (h)

Dynamometer : EC-70, Eddy Current

7.1 The results of Engine performance tests are tabulated in **Table-1** and graphically represented in **Fig. 4, 5 and 6**.

Table – 1

Power, (kW)	Crank shaft speed, (rpm)	Fuel consumption			Specific energy, (kWh/l)
		(l/h)	(kg/h)	Specific, (kg/kWh)	
1	2	3	4	5	6
Maximum power – 2 hours test:					
6.4	3603	2.50	2.04	0.321	2.54
6.0	3601	2.34	1.90	0.315	2.57*
Power at rated engine speed (3600 rpm):					
6.4	3603	2.50	2.04	0.321	2.54
6.0	3601	2.34	1.90	0.315	2.57*
Maximum torque:					
5.0	2250	1.49	1.22	0.246	3.32
4.9	2250	1.52	1.22	0.249	3.23*
Part loads:					
i) At torque corresponding to maximum power:					
6.4	3603	2.50	2.04	0.321	2.54
ii) 85% of the torque obtained in (i)					
5.5	3641	1.95	1.59	0.291	2.81
iii) 75% of the torque defined in (ii)					
4.1	3676	1.73	1.41	0.343	2.38
iv) 50% of the torque defined in (ii)					
2.8	3689	1.26	1.03	0.367	2.20
v) 25% of the torque defined in (ii)					
1.4	3710	1.04	0.85	0.622	1.31
vi) Unloaded					
--	3782	0.93	0.76	--	--

***High ambient conditions.**

Machine- 422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
------------------------------	---	-----------------------------

		<u>Natural ambient</u>	<u>High ambient</u>
- No load max. engine speed, (rpm)	:	3782	3790
- Crankshaft torque at max. Power, (Nm)	:	16.88	15.94
- Max. crankshaft torque, (Nm)	:	21.09	20.86
- Engine speed at max. crankshaft torque, (rpm)	:	2100 to 2250	2100 to 2250
- Back-up torque, (%)	:	24.94	---

Range of atmospheric conditions:

- Temperature (°C)	:	24 to 28	41 to 44
- Pressure, (kPa)	:	--	--
- Relative humidity, (%)	:	33 to 34	19 to 27

Max. temperatures (°C):

- Engine oil	:	120	128
- Fuel	:	29	44
- Air intake	:	30	44
- Exhaust gas	:	674	666

Pressure at max. power:

- Intake air, (kPa)	:	--	--
- Exhaust gas, (kPa)	:	3.1	2.1

Consumptions:

- Lub oil, (g/kWh)	:	--	0.95
--------------------	---	----	------

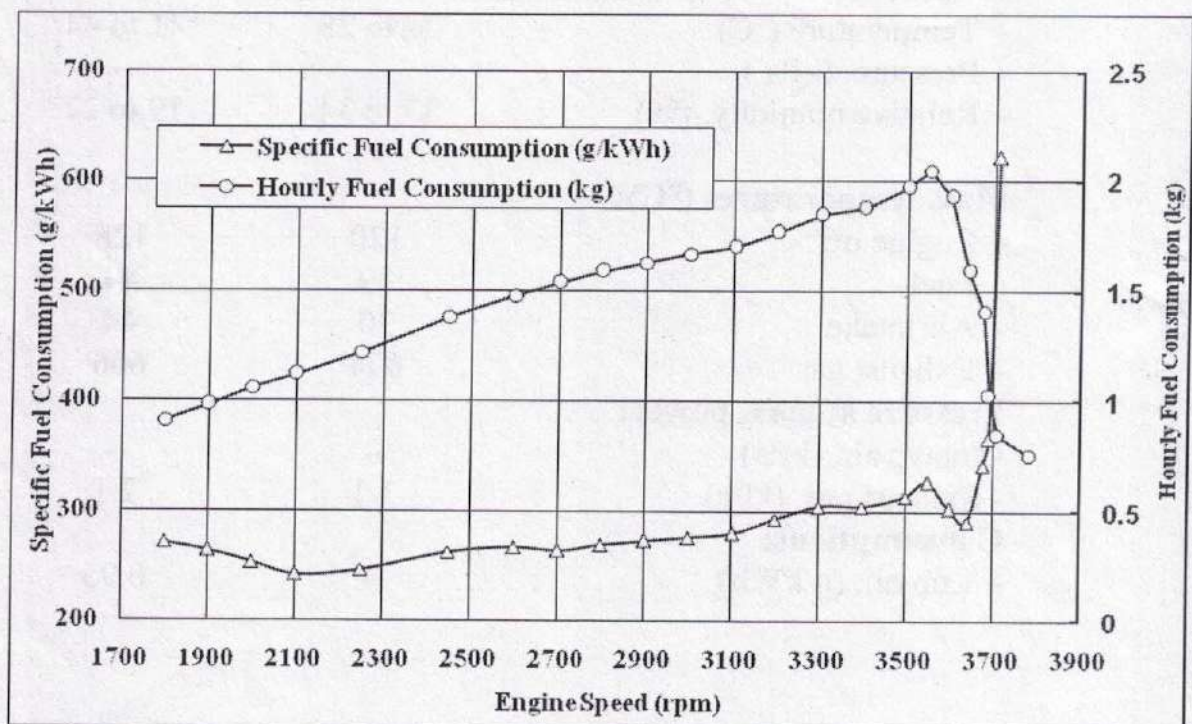
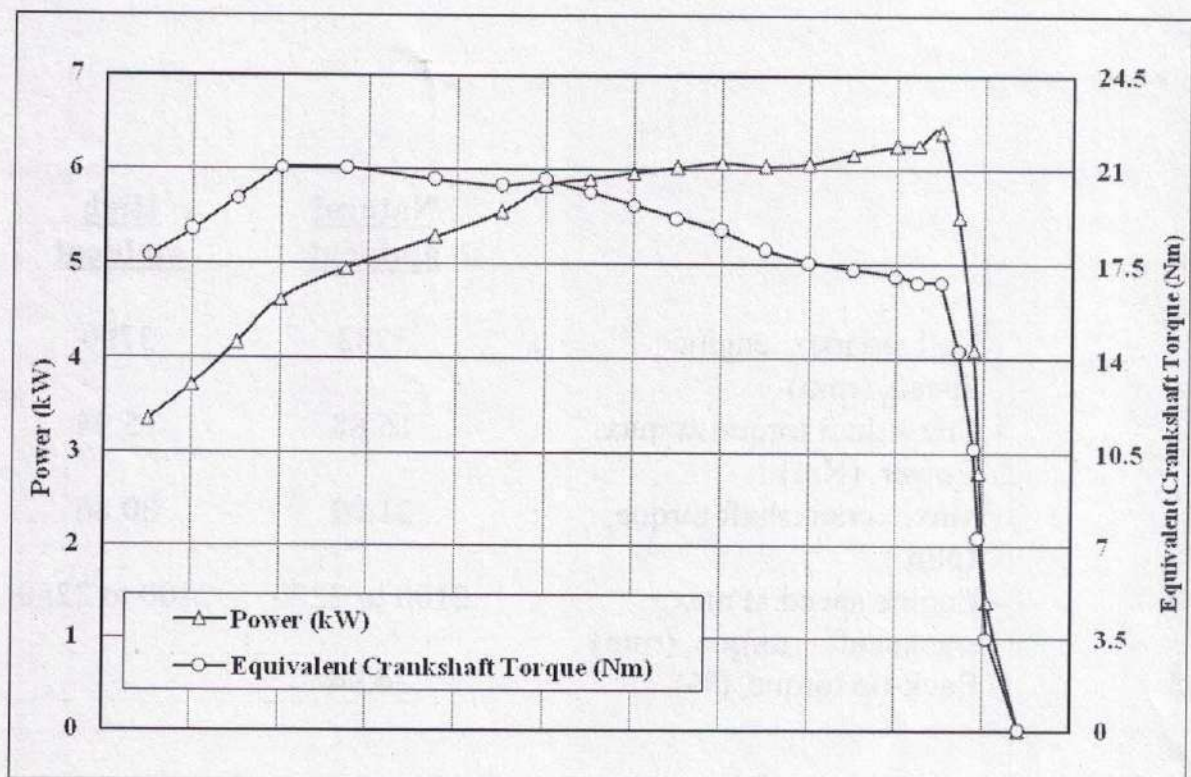


Fig. 4 ENGINE PERFORMANCE TEST (NATURAL AMBIENT)

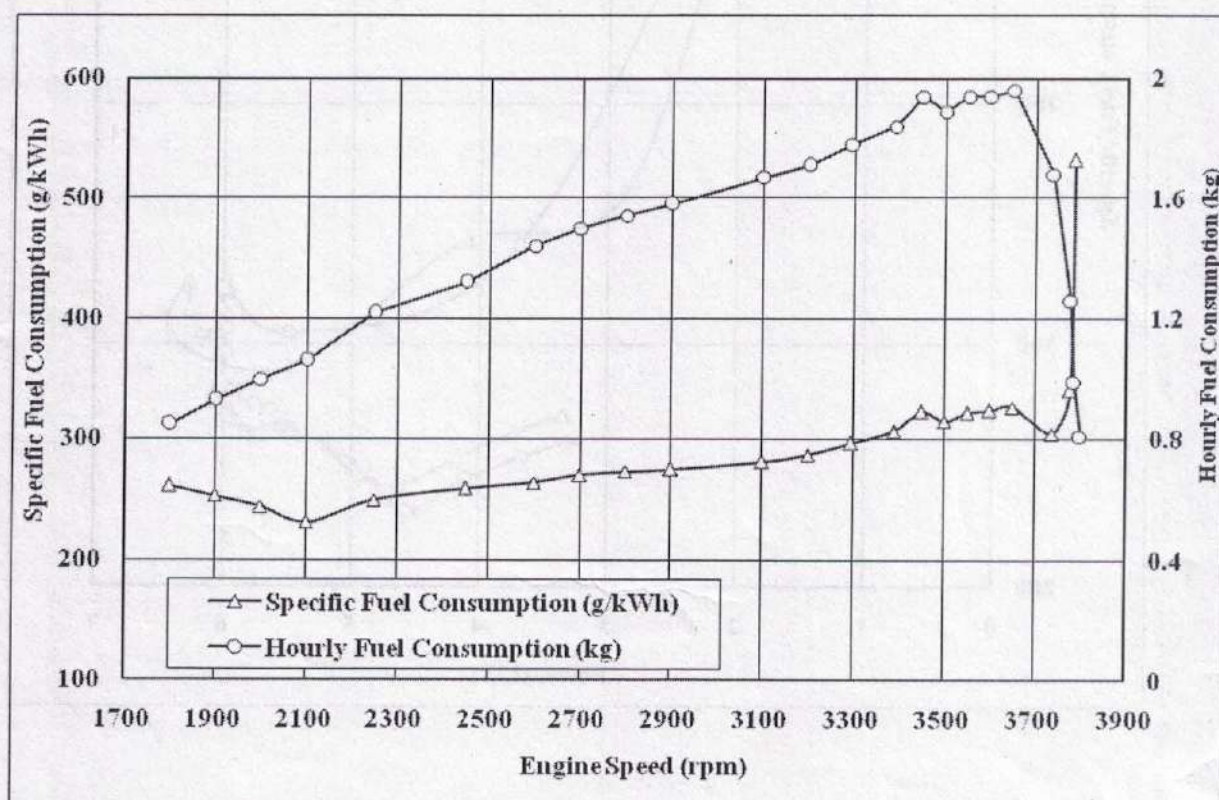
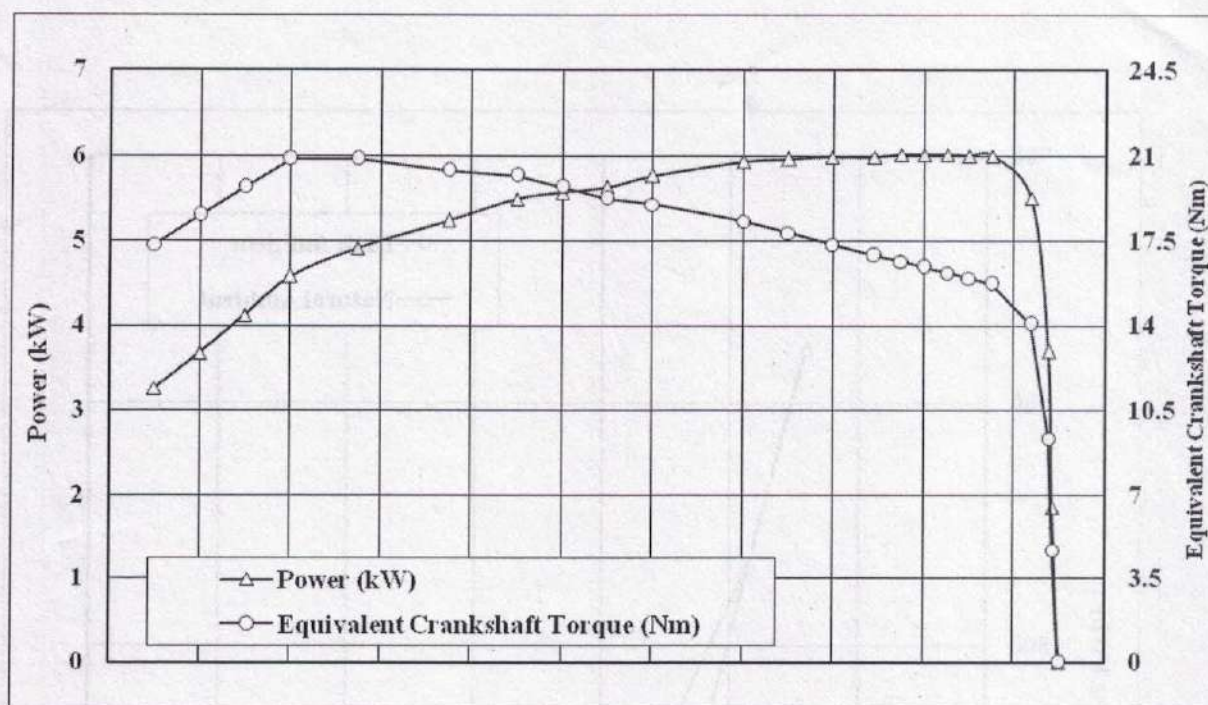


Fig. 5 ENGINE PERFORMANCE TEST (HIGH AMBIENT)

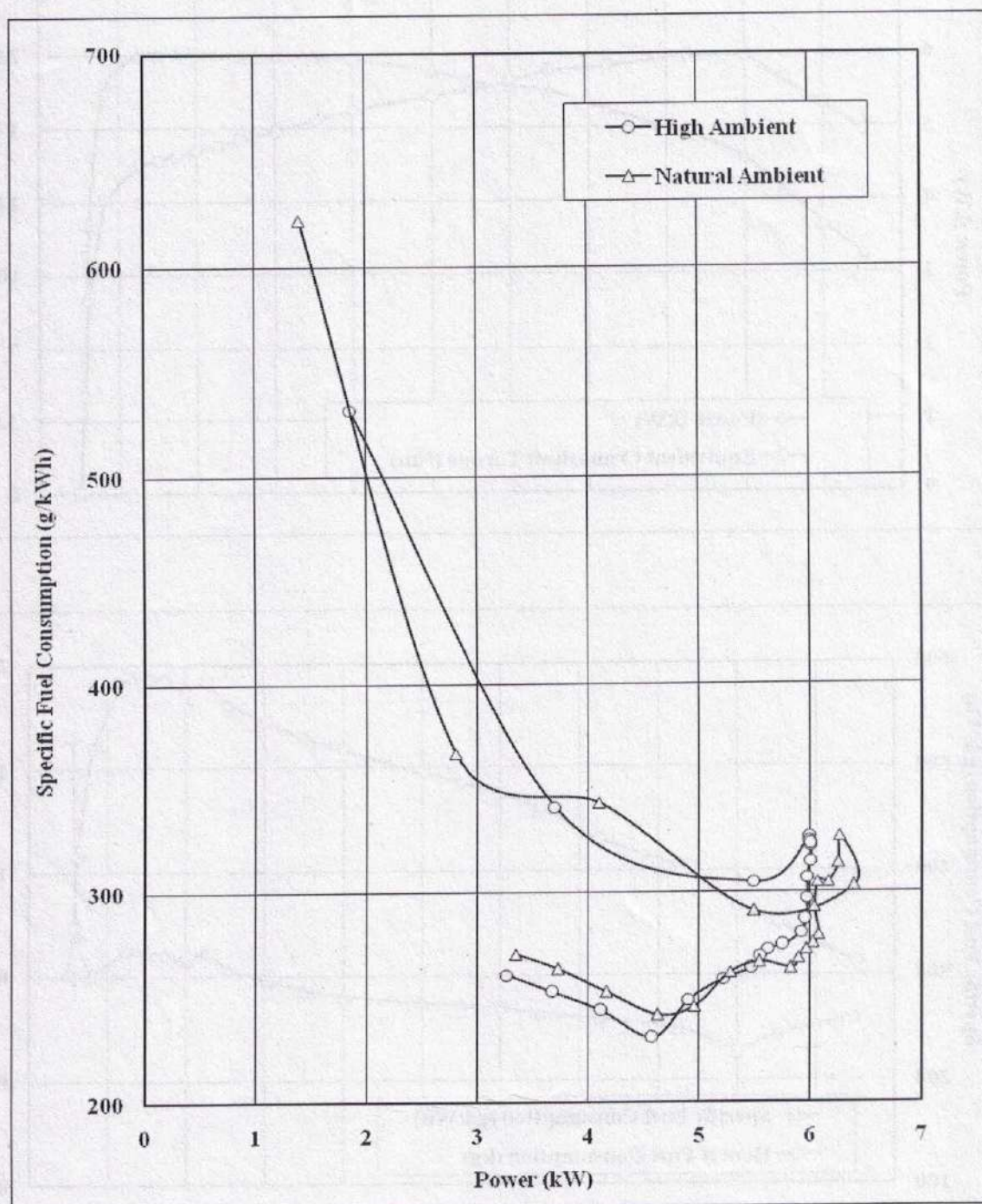


Fig. 6 ENGINE PERFORMANCE CHARACTERISTICS

Machine-422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
-------------------------	---	-----------------------------

8. AMPLITUDE OF MECHANICAL VIBRATION

Date of test : 22.02.2021
 Type of test surface : Concrete
 Type of instrument used : VM-31 MMF
 Test condition : At rated engine speed without load and all components are in working position.

Sl. No.	Location		X (μ)	Y (μ)
1.	Steering handle	Left	266*	284*
		Right	257*	264*
2.	Main clutch lever		292*	313*
3.	Fuel tank cap		258*	276*
4.	Fuel tank		275*	303*
5.	Gear shifting lever		251*	282*
6.	Rotary shield cover		296*	320*
7.	Tool bar		267*	292*
8.	Air cleaner		367*	410*
9.	Engine mounting base		462*	497*
10.	Silencer		597*	640*
11.	Accelerator lever		255*	299*
12.	Handle bar		267*	283

* The amplitude of Mechanical Vibration is on higher side

9. NOISE MEASUREMENT

Date of test : 22.02.2021
 Type of sound level meter : B&K, type 2239
 Type of test track : Loose Soil/Concrete
 Height of microphone, (m) : 1.2
 Test condition : At rated engine speed

Atmospheric conditions

Temperature, °C : 32.5
 Pressure, kPa : 97.5
 Relative humidity, % : 52.5
 Background noise level, dB(A) : 52
 Wind velocity, m/s : 1.2
 Observed noise level, dB(A)
 - By stander's position : 80
 - Operator's ear level : 95

Machine- 422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
------------------------------	---	-----------------------------

10. AIR CLEANER OIL PULLOVER TEST

Date of test : 09.02.2021
 Range of atmospheric conditions:
 Temperature, (°C) : 30.9 to 33.6
 Pressure, (kPa) : 97.1 to 97.3
 Relative humidity, (%) : 26 to 37
 Mass of oil before test, (g) : 35.8

Sl. No.	Position of power weeder	Loss of oil (g)	Oil pullover (%)
i)	Power Weeder parked on level ground	5.1	14.24
ii)	Power Weeder tilted 15° laterally on RHS	9.2	25.69
iii)	Power Weeder tilted 15° laterally on LHS	1.6	4.47
iv)	Power Weeder tilted 15° longitudinally with front end up	1.2	3.35
v)	Power Weeder tilted 15° longitudinally with rear end up	1.5	4.19

11. TURNING ABILITY

Date of test : 10.02.2021
 Type of test track : Concrete

The minimum turning diameter and the minimum clearance diameter of the machine was measured and recorded as under:

Characteristics	Minimum turning diameter, m	Minimum clearance diameter, m
LHS	1.48	3.02
RHS	1.52	2.98

Machine- 422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
------------------------------	---	-----------------------------

12. HARDNESS AND CHEMICAL COMPOSITION

12.1 Hardness of rotor blades :

The surface hardness of blade was recorded as under: -

Sl. No.	As per IS 6690:1981 (Reaffirmed 2012)	As observed (HRC)	Remarks
1	56 ±3 HRC	56.0	Conforms
2		54.6	
3		54.0	
4		57.0	
5		55.0	

12.2 Chemical composition of rotor blades :

The material of rotary blade was got analyzed from Geological and Metallurgical Laboratories, Bangalore for chemical composition. The results of chemical analysis test results vide test report No. GML 5946-5/2020-2021, dated 17.02.2021 is as under:

Constituents	As per IS 6690:1981 (Reaffirmed 2012)		Composition as observed (% of weight)	Remarks
	Carbon Steel	Silicon Manganese Steel		
Carbon (C)	0.70 -0.85	0.50-0.60	0.578	Conforms
Silicon (Si)	0.10 -0.40	1.50-2.00	1.922	Conforms
Manganese (Mn)	0.50 -1.0	0.50-1.00	0.744	Conforms
Sulphur (S)	0.05(max)	0.05(max)	0.006	Conforms
Phosphorous (P)	0.05(max)	0.05(max)	0.028	Conforms
Boron (B)	---	---	< 0.001	Conforms

Machine- 422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
------------------------------	---	-----------------------------

13. FIELD PERFORMANCE TEST

The field tests were conducted for 26.0 hours for testing the said Power Weeder. During the field tests no load engine speed adjusted to 3650 ± 5 rpm. In all, five tests were conducted in medium soil. The detailed test results are represented in the Annexure and summarized in the ensuing table:

Sl.No.	Parameters	Range
1	Type of soil	Medium
2	Av. soil moisture (%)	7.2 to 12.8
3	Av. bulk density of soil (g/cc)	1.55 to 1.87
4	Av. speed of operation (kmph)	1.50 to 1.53
5	Av. depth of cut (cm)	7.0 to 7.3
6	Av. width of cut (cm)	143.0 to 145.0
7	Av. area covered (ha/h)	0.144 to 0.149
8	Av. time required for one ha (h)	6.71 to 6.94
9	Av. field efficiency (%)	65.75 to 68.66
10	Av. weeding efficiency (%)	59.67 to 65.69
11	Av. fuel consumption	
	- l/h	0.90 to 1.50
	- l/ha	6.17 to 10.41

13.1 Rate of work

- Av. rate of work was recorded as 0.144 to 0.149 ha/h and the speed of operation vary from 1.50 to 1.53 kmph.
- Av. time required to cover one hectare was recorded as 6.71 to 6.94.

13.2 Quality of work

- Av. depth of cut was recorded as 7.0 to 7.3 cm.
- Av. working width was observed as 143.0 to 145.0 cm.
- Av. field efficiency was found as 65.75 to 68.66 %.
- Av. weeding efficiency was found as 59.67 to 65.69 %.

13.3 Adequacy of power of prime mover

The power of prime mover was found adequate.

Machine- 422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
------------------------------	---	-----------------------------

13.4 Wear analysis of rotor blades

Sl. No.	Initial mass(g)	Final mass (g)	Loss of mass (g)	Percentage of wear	
				After 26 h	Per hour
1	322.9	310.9	12.0	3.72	0.14
2	309.3	297.5	11.8	3.81	0.15
3	320.6	314.2	6.4	1.99	0.08
4	310.2	300.3	9.9	3.19	0.12
5	318.1	308.0	10.1	3.17	0.12
6	322.7	313.8	8.9	2.76	0.11
7	333.9	324.0	9.9	2.96	0.11
8	318.6	310.9	7.7	2.42	0.09
9	327.8	316.3	11.5	3.51	0.13
10	332.5	325.0	7.5	2.25	0.09

14. EASE OF OPERATION & ADJUSTMENTS

No noticeable difficulty was observed during the operation & adjustment of the machine.

15. DEFECTS, BREAKDOWNS AND REPAIRS

No defects & breakdowns was observed during the entire test.

16. COMPONENTS / ASSEMBLY INSPECTION

The Power Weeder was dismantled after 46.27 hours of operation at this Institute.

16.1 Engine:

Cylinder bore:

Cylinder No.	Cylinder bore dia, (mm)						Max. permissible wear limit, (mm)
	Top position		Middle position		Bottom position		
	Thrust Side	Non- thrust Side	Thrust side	Non- thrust Side	Thrust Side	Non- thrust side	
1.	86.03	86.02	86.03	86.02	86.03	86.03	86.18

Machine- 422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
----------------------	-------------------------------------	---------------------

Piston:

Piston No.	Piston dia (mm)				Clearance between piston & cylinder liner at the skirt of the piston, (mm)	Maximum permissible clearance limit, (mm)
	Top(above top compression ring)		At skirt			
	Thrust side	Non- thrust Side	Thrust side	Non- thrust side		
1.	85.40	85.31	85.65	***	0.09	0.30

*Not measured due to piston design constraints

Ring end gap:

Rings	Ring end gap, (mm)			Max. permissible end gap limit, (mm)
	Top	Middle	Bottom	
1 st comp. ring	0.33	0.33	0.32	1.00
2 nd comp. ring	0.47	0.45	0.45	
Oil ring	0.35	0.35	0.36	

Ring side clearance:

Rings	Ring side clearance, (mm)	Max. permissible clearance limit, (mm)
1 st Compression ring	0.08	0.20
2 nd Compression ring	0.08	0.18
Oil ring	0.04	0.17

Main and big end bearings:

Main bearings:

Bearing No.	Type of bearings	Diametrical clearance, (mm)	Crankshaft end float, (mm)	Max. permissible clearance limit,(mm)	
				Diametrical clearance	Crankshaft end float
1.	Bush bearing (timing gears side)	0.04	0.30	Not specified	0.25
2.	Ball bearing (Flywheel side)	NA			

Machine- 422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
------------------------------	---	-----------------------------

Big end bearings:

Bearing No.	Clearance, (mm)		Max. permissible clearance limit,(mm)	
	Diametrical	Axial	Diametrical	Axial
1.	0.039	0.20	0.033	0.3

Valve, guides and timing gears:

Observation

Any marked sign of : None

overheating of valves

Pitting of seat/faces of : None

valves

Any visual damage to the : None

teeth of timing gears

Clearance between valve :
guide and valve stem
(mm)

- Intake valve : 0.07

- Exhaust valve : 0.08

16.2

Clutch:

Any marked wear in : None

clutch friction plate

Thickness of clutch : 3.4

plate, (mm)

Condition of clutch : Normal

release bearing.

Condition of pilot : Normal

bearing

Condition of pressure : Normal

plate

Presence of oil and : None

water in clutch housing

16.3

Transmission gears:

All the gears of the transmission system were found in normal condition.

16.4

Rotary drive unit:

The rotary drive unit was dismantled and all the components were found in normal condition.

Machine- 422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
------------------------------	---	-----------------------------

17. COMMENTS & RECOMMENDATIONS

17.1 Engine Performance:

- 17.1.1 The maximum power and rated power were observed 6.4 kW under natural ambient condition against the declared value of 6.9 kW & 6.3 kW respectively.
- 17.1.2 Specific fuel consumption of engine corresponding to maximum power was recorded as 321 g/kWh against the declared value of 300 g/kWh.
- 17.1.3 Max. Torque was observed as 21.09 Nm against the declared value of 18.7 Nm.
- 17.1.4 Back up torque of engine was observed as 24.94 %, against the declared value of 12 %.
- 17.2 Max. Noise at operator's ear level & by stander's position was observed as 95 & 80 dB (A) respectively against the limit specified by the ILO.
- 17.3 The amplitude of mechanical vibration on most of the assemblies of the Power Weeder was observed up to the maximum extent of 640 microns, which is on higher side. Therefore necessary action to dampen the vibration should be taken in to account.
- 17.4 The hardness of rotary blades conforms to the requirement of IS 6690:1981 (Reaffirmed 2012).
- 17.5 The chemical composition of rotary blades conforms to the requirement of IS 6690:1981 (Reaffirmed 2012).
- 17.6 The provided labeling plate should be as per the requirement IS.
- 17.7 The maximum power and rated power declared by manufacturer/ applicant is 6.9 kW and 6.3 kW respectively. However on labeling plate of said power weeder it is mentioned as 5.9 kW @3600 rpm. This should be looked into.
- 17.8 The crankshaft end float is observed as 0.30 mm against declared value of 0.25 mm. This is on higher side against declared value and should be looked into.

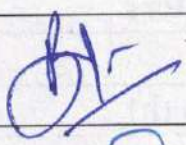
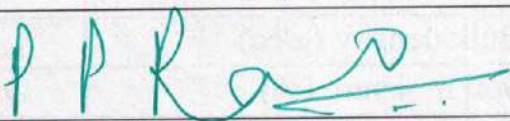
Machine- 422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
------------------------------	---	-----------------------------

17.9 Technical literature:

Operation manual, service manual and parts catalogue power weeder and engine were supplied with the test sample for reference during the test. It is however, recommended that same may be revised and brought out in Hindi & other regional languages as per IS 8132:1999 (Reaffirmed 2004) for the sake of user & technical personnel.

The adequate literatures are required to supply in booklet form.

TESTING AUTHORITY:

B. N. DIXIT AGRICULTURAL ENGINEER	
P. C. MESHARAM SENIOR AGRICULTURAL ENGINEER	
Dr. P. P. RAO DIRECTOR	

18. APPLICANT'S COMMENTS

Para No.	Our reference	Applicant's comments
18.1	17.1.2	We will work on comments for better result of Fuel Consumption, In support with our R & D and Production Team. We will make all possible necessary changes in our product in future.
18.2	17.2	In future we will give positive efforts to decrease further and maintain standards which are specified by the ILO. We will make all possible necessary changes in our product in future.
18.3	17.3	Noted, We will work on it and take necessary steps, in support with our R & D and Production team. We will provide anti vibration rubber grips at handle bars for operator comfort, safety and health. We will also look into adding vibration damping gaskets at all joint wherever possible. We will make all possible necessary changes in our product in future
18.4	17.6	We will provide as per the IS. We will make all possible necessary changes in our product in future
18.5	17.7	Noted, It was not harmful to user because actual power observed during the test is higher against value of power printed on name plate. It was consider as "Under Promise Over Delivery" with our promise of "Honour customer's Trust" as per company tag line. We will look on comment and update it on future Production after Final Report. We will make all possible necessary changes in our product in future
18.6	17.8	Noted, We will work on it and take necessary steps, In support with our R & D and Production Team. We will make all possible necessary changes in our product in future.
18.7	17.9	Yes, We provided English & Hindi Language Operation manual, Service manual and Part catalogue of Power Weeder and Engine During the test. We will also look into adding Multiple Language Literature as per IS 8132: 1999(Reaffirmed 2004).

Machine- 422/1256	FARMWORLD FWMT420DE POWER WEEDER	COMMERCIAL (ICT)
------------------------------	---	-----------------------------

ANNEXURE

FIELD PERFORMANCE RESULTS

Place of test: Institute Farm Fields, Anantapur Dist. A.P.

Sl. No.	Parameters	I	II	III	IV	V
1	Date of test	11.02.21	12.02.21	13.02.21	14.02.21	15.02.21
2	Net test duration (h)	5.00	5.00	7.00	5.00	4.00
3	Furrow length (m)	93.0	85.0	90.0	90.0	88.0
4	Type of soil	Medium				
5	Bulk density (g/cc)	1.55	1.84	1.87	1.82	1.80
6	Soil moisture (%)	7.2	12.3	12.8	12.1	11.8
7	Previous treatment	Nil				
8	Forward speed, (kmph)	1.53	1.50	1.51	1.51	1.51
9	Av. depth (cm)	7.1	7.2	7.3	7.0	7.1
10	Av. width (cm)	143.1	143.0	144.0	145.0	144.0
11	Area covered (ha/h)	0.146	0.144	0.144	0.144	0.149
12	Time required for one ha (h)	6.85	6.94	6.94	6.94	6.71
13	Field efficiency (%)	66.67	67.29	66.36	65.75	68.66
14	Av. height of weeds, (cm)	13.40	15.20	12.60	16.60	16.00
15	Av. number of weeds per m ² (Before operation)	60.33	80.67	61.33	63.00	73.00
16	Av. number of weeds per m ² (After operation)	24.33	27.67	22.33	24.00	26.33
17	Weeding efficiency (%)	59.67	65.69	63.59	61.90	63.93
18	Fuel consumption					
	- l/h	1.26	1.50	1.30	0.90	0.92
	- l/ha	8.63	10.41	9.02	6.25	6.17



भारतसरकार

Government of India

कृषि एवं किसान कल्याण मंत्रालय

Ministry of Agriculture and Farmers Welfare

कृषि एवं किसान कल्याण विभाग

Department of Agriculture and Farmers Welfare

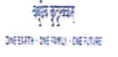
दक्षिणी क्षेत्र कृषि यंत्र प्रशिक्षण एवं परीक्षण संस्थान,

Southern Region Farm Machinery Training & Testing Institute

ट्रैक्टर नगर, गार्लदिन्ने, जि. अनंतपुर (आं. प्र.)

Tractor Nagar, Garladinne, District: Anantapur (A.P.)- 515731

An I.S.O. 9001- 2015 Certified Institute

Web.: <http://srfmtti.dacnet.nic.in>E-mail: fmti-sr@nic.in

File No. 16-99/2023-24-Test

Dated: 30.07.2024

To

M/s. FarmworldAgrotech Pvt. Ltd.,
SF 08, Shreemad Bhavan,
Opp. KantaStree Vikas Gruh, Dhebar Road,
Rajkot – 360 002, Gujarat, India.

Subject:-Extension of validity ICT tested Power Weeder– regarding.

Sir,

With reference to your email dated 27.12.2023 regarding the subject cited above, it is to state that your tested samples come under the category of **Self Propelled machinery** that have seven years of validity for ICT test as per Ministry's letter No.13-22/2020 -M&T (I&P) dated 12th December, 2023.

Sl. No	Name of the Machine	Make & Model	Type of Test	Test Report Number	Report Release Date	Validity of Test Report	Extended date of Validity
1	Power Weeder	FARMWORLD, FWMT420DE	ICT	Machine-422/1256	March, 2021	28.02.2026	28.02.2028
2	Power Weeder	FARMWORLD FWMT 210P	ICT	Machine-426/1264	March, 2021	28.02.2026	28.02.2028

(Pramod Yadav)
Agricultural Engineer
For Director

Copy to :

1. Applicant copy.
2. Correspondence file.

कृषि अभियंता /Agricultural Engineer
भारत सरकार/Government of India
द क्षे.का.म.प्र.एव.परी.संस्थान/S.R.F.M.T.&T.I
गार्लदिन्ने/GARLADINNE-515731
अनंतपुर जिला.(आं.प्र.)/ANANTAPUR DIST. (A.P.)